

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (16 marks – approximately 29 minutes)

- (a) Define the term "omni-channel" in e-commerce. (2 marks)
- (b) Explain why Hong Kong was described as an e-commerce "wasteland". (4 marks)
- (c) Justify the importance of the following concerns in designing a new e-commerce platform:
- (i) Identity authentication among the customers; and (3 marks)
 - (ii) Digital communication security between the customers and the e-commerce platform. (3 marks)
- (d) List and describe any TWO applications of an e-business system. (4 marks)

Question 2 (16 marks – approximately 29 minutes)

(a) Classify each of the following application controls as a/ an input, process or output control:

(i) Validity test; (1 mark)

(ii) Echo check; (1 mark)

(iii) Closed-loop verification; (1 mark)

(iv) Spooling; (1 mark)

(v) Record locking; and (1 mark)

(vi) Parity check. (1 mark)

(b) Contrast any ONE difference between black hat hackers and white hat hackers. (4 marks)

(c) According to a cyber security report in 2018, there were over 400,000 distributed denial of service ("DDoS") attacks every month worldwide. Justify the effectiveness of each of the following cyber security measures to block DDoS attacks:

(i) Utilising intrusion prevention systems; and (3 marks)

(ii) Using firewalls. (3 marks)

Question 3 (20 marks – approximately 36 minutes)

- (a) State and describe any TWO decision-making techniques that could be performed by a decision support system.
(6 marks)
- (b) Demonstrate how we can use information systems to facilitate performance management activities in sales and marketing.
(5 marks)
- (c) Explain the necessity of establishing a code of ethical conduct on using information systems.
(5 marks)
- (d) State and describe any TWO relevant standards relating to IT security that auditors should be familiar with.
(4 marks)

Question 4 (16 marks – approximately 29 minutes)

- (a) State and explain any TWO responsibilities of a data owner. (4 marks)
- (b) Imagine that your boss is now exploring some solutions to store electronic files and documents at a centralised location via an internal network. Contrast the difference between a storage area network ("SAN") and a network attached storage ("NAS"). (4 marks)
- (c) List any TWO benefits of using an extranet in supply chain management. (4 marks)
- (d) Outline any TWO examples of prediction outcomes that could be generated from Big Data. (4 marks)

Question 5 (12 marks – approximately 21 minutes)

- (a) Illustrate how an accountant serves as a subject expert in a system development project.
(4 marks)
- (b) Users of a new accounting system are normally referred as "change agents", and some of them may resist accepting and using the new system. Use any TWO techniques for addressing resistance to a new accounting system.
(4 marks)
- (c) Define "change management" in the context of information systems. Explain why the change management process could be considered a quality assurance process.
(4 marks)

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